

Parameter-Efficient Multilingual Adaptation: A Comparative Study of GPT-2 and Qwen2-1.5B on XNLI

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Abstract

Full-parameter fine-tuning (FT) enables strong task performance in pretrained language models but scales poorly with model size and offers limited reusability across tasks and languages. This work investigates Low-Rank Adaptation (LoRA) as a parameter-efficient alternative for Natural Language Inference (NLI) in both monolingual and multilingual settings. Building on the baselines established in Tasks 2–3, we first show that GPT-2 LoRA matches the performance of full FT on English XNLI while updating fewer than 1% of parameters. We then evaluate multilingual LoRA on GPT-2 and find that, although it reduces the interference observed in multilingual full FT, overall cross-lingual gains remain limited by GPT-2’s English-only pretraining. To assess the role of multilingual model priors, we apply LoRA to Qwen2-1.5B, a modern multilingual LLM. English-only LoRA yields strong zero-shot performance across 15 languages, and multilingual LoRA further improves accuracy—especially for lower-resource languages such as Swahili—while preserving high English performance. Across all experiments, results show that LoRA provides an efficient and robust adaptation mechanism, but that the underlying multilingual pretraining, rather than the fine-tuning strategy, is the dominant factor shaping cross-lingual generalisation.

1 Introduction

Large language models (LLMs) such as GPT-2 (Radford et al., 2019) achieve strong performance on many natural language understanding tasks under full-parameter fine-tuning (FT). However, full FT scales poorly: updating all weights requires substantial GPU memory, increases compute cost, and makes it expensive to maintain separate models for multiple tasks or languages. These challenges motivate the study of parameter-efficient fine-tuning (PEFT) methods that reduce training cost while preserving model quality.

In Tasks 2 and 3 of this project, we established two baselines. First, GPT-2 full FT performs well on English XNLI (Conneau et al., 2018) but provides limited zero-shot transfer to other languages. Second, multilingual FT on GPT-2 yields only moderate cross-lingual gains and introduces interference, reflecting structural limits imposed by GPT-2’s English-only pretraining.

This work investigates Low-Rank Adaptation (LoRA) (Hu et al., 2021) as a parameter-efficient alternative to full FT. LoRA injects small, trainable low-rank matrices into selected linear projections while freezing the original weights, enabling task adaptation while updating less than 1% of parameters.

We conduct four extension experiments to evaluate LoRA across model scales and multilingual settings. We first test whether GPT-2 LoRA can match full FT on English NLI. Next, we train a multilingual GPT-2 LoRA model on five languages to examine whether PEFT can mitigate the interference observed in full FT. We then apply LoRA to Qwen2-1.5B, a modern multilingual model, to study its zero-shot transfer across 15 languages. Finally, we fine-tune Qwen2-1.5B LoRA on multiple languages to assess how PEFT interacts with multilingual pretraining.

Together, these experiments provide insight into how parameter-efficiency, model scale, and multilingual pretraining jointly influence cross-lingual generalisation, addressing the limitations observed in Tasks 2 and 3.

2 Background

2.1 Natural Language Inference

Natural Language Inference (NLI) requires a model to determine whether a hypothesis is *entailed by*, *contradicted by*, or *neutral with respect to* a given premise. The XNLI dataset (Conneau et al., 2018) extends this task to 15 languages, making it a stan-

standard benchmark for studying cross-lingual transfer. Because NLI involves reasoning over paired sentences, it is a strong diagnostic task for how well models adapt their representations during fine-tuning.

2.2 Full-Parameter Fine-Tuning

Full fine-tuning updates all weights of a pretrained model, allowing maximal task adaptation. While effective for smaller models such as GPT-2, it scales poorly: memory usage grows with parameter count, training becomes expensive, and separate fine-tuned models must be stored for each task or language. In Tasks 2 and 3, GPT-2 full FT achieved strong English performance but showed limited multilingual transfer and interference when multiple languages were trained jointly.

2.3 Parameter-Efficient Fine-Tuning and LoRA

Parameter-efficient fine-tuning (PEFT) aims to adapt models while updating only a small portion of parameters. LoRA inserts low-rank trainable matrices into selected linear projections—typically attention and feed-forward layers—while freezing the original weights (Hu et al., 2021). This reduces trainable parameters by over 99% and lowers memory cost, enabling efficient multilingual and multi-task training. In this work, LoRA serves as the primary alternative to full FT.

2.4 Multilingual Pretraining and Cross-Lingual Transfer

GPT-2 was pretrained solely on English text (Radford et al., 2019), limiting its ability to represent other languages. This is reflected in high tokenizer fertility and poor zero-shot performance in Task 3. Modern multilingual models such as Qwen2-1.5B, trained on diverse corpora, provide stronger cross-lingual priors. Comparing GPT-2 and Qwen2 allows us to isolate the effects of multilingual pretraining when applying PEFT.

Taken together, these considerations motivate our extension experiments, which compare LoRA-based adaptation across two model classes (GPT-2 vs. Qwen2-1.5B) and across English-only, multilingual, and zero-shot evaluation settings.

3 Methodology

Tasks 1–3 were executed on Kaggle and Google Colab using an NVIDIA T4 GPU (16 GB VRAM). Extension experiments involving Qwen2-1.5B used

Runpod with RTX 6000 ADA due to higher compute required.

Datasets were sourced from the HuggingFace NLI collection and the official TSV releases (Conneau et al., 2018). Tokenization across all tasks used the GPT-2 BPE tokenizer (Radford et al., 2019).

3.1 Task 1 — Reconstruction of GPT-2 and AdamW

Task 1 re-implemented the core components of GPT-2 including embeddings, causal self-attention, feed-forward layers, residual connections, and tied output projections—along with a AdamW optimizer (Loshchilov and Hutter, 2019). We verified correctness through numerical checks against the HuggingFace model and a reference optimizer.

To validate the full training pipeline, we pre-trained the toy model on a small English corpus (5,778 lines) using next-token prediction. Over three epochs, the loss decreased smoothly, confirming that the architecture, optimizer, and data flow were implemented correctly.

3.2 Task 2 — English NLI Fine-Tuning

Task 2 established the supervised baseline for the NLI task. We framed NLI as a generative next-token prediction problem by concatenating the premise, hypothesis, and label fields into a single text prompt:

Premise: <premise_text>
Hypothesis: <hypothesis_text>
Label: (“entailment”, “neutral”, or “contradictory”)

During training, the model was trained to generate the correct label by minimizing the negative log-likelihood of the target label token.

Table 1 summarizes the hyperparameters used for the English NLI fine-tuning task. Because GPT-2 is highly sensitive to learning rate, we conducted a small LR sweep which is reported in the Results section.

Hyperparameter	Value
Weight Decay	1×10^{-2}
Batch Size	4
Epochs	1
Sequence Length	1024
Learning Rate	Variable (See Table 3)

Table 1: Configuration for Task 2

3.3 Task 3 — Multilingual NLI

Task 3 evaluated GPT-2’s multilingual capabilities and informed our selection of languages for further fine-tuning.

First, we evaluated the English-fine-tuned model from Task 2 on all 15 XNLI languages in a zero-shot setting. To explain the large variation in performance, we computed tokenizer fertility—the average number of subword tokens per word—for each language. In Table 2, we observed a clear negative correlation between fertility and accuracy, indicating that GPT-2’s tokenizer provides uneven support across scripts and languages.

Based on this analysis, we selected Spanish, French, German, and Swahili (plus English) for multilingual fine-tuning. We then trained both per-language models and a unified multilingual model to compare isolated vs. shared supervision.

Language	Fertility	Zero-Shot Acc (%)
<i>Selected for Training</i>		
English (en)	1.11	78.34
French (fr)	1.75	41.74
Spanish (es)	1.83	46.87
Swahili (sw)	2.08	38.54
German (de)	2.10	40.28
<i>Other Languages</i>		
Turkish (tr)	2.73	39.82
Vietnamese (vi)	3.62	37.56
Chinese (zh)	3.77	38.44
Arabic (ar)	4.70	35.91
Hindi (hi)	5.12	34.73
Urdu (ur)	5.16	34.63
Bulgarian (bg)	5.53	35.17
Russian (ru)	5.90	35.09
Greek (el)	6.16	36.89
Thai (th)	9.48	34.45

Table 2: Tokenizer fertility and zero-shot accuracy. Languages with lower fertility (< 2.5) were selected for multilingual fine-tuning experiments.

3.4 Extension 1 - English NLI LoRA for GPT-2

Task 2 showed that GPT-2 full fine-tuning achieves strong performance in English NLI. In this extension, we replace full-parameter fine-tuning with LoRA and update only the attention projection, feedforward expansion, and feedforward projection modules. LoRA configuration: rank = 8, α = 16, dropout = 0.1, updating 1.18M parameters (0.94%).

3.5 Extension 2: Multilingual NLI LoRA for GPT-2

Task 3 demonstrated that multilingual full fine-tuning on GPT-2 yields modest gains but suffers from English performance degradation and cross-lingual interference. Here, we examine whether the same training budget—250,000 optimisation steps—can be used more efficiently via LoRA.

We train a single GPT-2 LoRA model on a unified mixture of five languages: **English, German, Spanish, French, Swahili** to match Task 3. However, minor differences in preprocessing, shuffling, and training framework prevent a strict one-to-one comparison explained in limitations.

3.6 Extension 3: LoRA Fine-Tuning Qwen2–1.5B on English NLI

GPT-2 provides limited multilingual capability due to its English-only pretraining. To evaluate the effect of model scale and multilingual pretraining, we apply LoRA to **Qwen2–1.5B**, a modern multilingual LLM.

We train Qwen2–1.5B LoRA on the complete English NLI dataset using the same prompting format as in Tasks 2–3. After training, we perform zero-shot evaluation on all 15 XNLI languages to quantify cross-lingual generalisation.

This extension isolates the role of pretraining corpus and model capability.

3.7 Extension 4: Multilingual LoRA on Qwen2–1.5B

Finally, we extend LoRA training on Qwen2–1.5B to a multilingual setting. We fine-tune the model on a concatenation of five languages—{en, de, es, fr, sw}—using 50% of each training set for computational feasibility.

This experiment mirrors Extension 2 but on a much more capable base model, allowing comparison across:

- English-only LoRA (Qwen2),
- Unified multilingual LoRA (Qwen2),
- Unified multilingual LoRA (GPT-2),
- Full multilingual FT (GPT-2 from Task 3).

Together, these four extensions illustrate how LoRA interacts with model scale, language coverage, and training regimes.

4 Results & Analysis

4.1 Task 1 Results

We trained a small GPT-2 model (13.2M parameters, 2 layers) on a toy corpus of 5,778 lines for three epochs to verify the correctness of our implementation. The training loss decreased smoothly from 10.86 to 6.76 (Figure 1), corresponding to a perplexity of about 860. Most of the improvement occurred in the first epoch, with later epochs providing incremental refinement.

The stable convergence without oscillation indicates that key architectural components—causal masking, residual connections, layer normalization, and our AdamW implementation—were functioning correctly. Despite the limited dataset and reduced model size, the model demonstrated the expected behaviour of early-stage language pre-training, validating our end-to-end GPT-2 implementation before applying it to downstream NLI tasks.

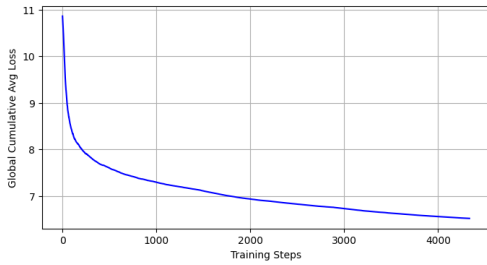


Figure 1: Training loss curve for Task 1. Loss decreases steadily, confirming correct implementation.

4.2 Task 2 Results

Table 3 shows the development and test accuracy of the baseline model and several hyperparameter variations.

The baseline model achieved approximately 78–80% accuracy on the English test set. Training was stable and accuracy improved consistently with learning-rate tuning, with the best configuration reaching 81.50%.

Overall, the results confirm that GPT-2 adapts well to English NLI with minimal training: the model requires only one epoch of supervised fine-tuning to reach strong performance.

4.3 Task 3 Results

We evaluated GPT-2 in three multilingual settings: (i) zero-shot transfer from the English-only model, (ii) per-language fine-tuning, and (iii) unified multilingual fine-tuning on five selected languages (en,

Model Config	Learning Rate	Avg Loss	Accuracy (%)
1 (Baseline)	5×10^{-5}	0.6139	78.58
2	5×10^{-4}	1.0005	52.24
3	1×10^{-6}	0.7662	76.33
4 (Best)	2×10^{-5}	0.5820	81.50

Table 3: Impact of Learning Rate on Test Accuracy

de, es, fr, sw), chosen based on zero-shot performance and tokenizer fertility.

Table 4 shows the results. Per-language models achieved the highest accuracy for their respective languages, while the unified multilingual model—trained on a 250k sampled subset—still provided substantial gains over zero-shot across all five languages. English accuracy decreased (78.34% $\rightarrow$ 72.30%), reflecting cross-lingual interference in the shared model.

The unified model also improved performance on unseen languages such as Turkish, Thai, and Vietnamese by 3–5 points compared to the English-only zero-shot baseline. This indicates that even limited multilingual supervision helps GPT-2 develop more general cross-lingual representations, despite its fundamentally monolingual pretraining.

Language	Zero-Shot	Per-Lang	Unified
<i>In Training Set</i>			
English (en)	78.34	78.34	72.30
Spanish (es)	46.87	66.49	60.58
French (fr)	41.74	63.93	59.42
German (de)	40.28	65.39	59.34
Swahili (sw)	38.54	57.96	53.77
<i>Unseen</i>			
Turkish (tr)	39.82	–	43.39
Chinese (zh)	38.44	–	41.00
Vietnamese (vi)	37.56	–	40.38
Greek (el)	36.89	–	39.44
Thai (th)	34.45	–	38.96
Hindi (hi)	34.73	–	38.46
Bulgarian (bg)	35.17	–	37.92
Russian (ru)	35.09	–	37.15
Arabic (ar)	35.91	–	35.75
Urdu (ur)	34.63	–	35.73

Table 4: Accuracy (%) for zero-shot, per-language, and unified multilingual GPT-2 models.

4.4 Extension 1 — English GPT-2 LoRA vs. Full FT

Table 5 shows that LoRA achieves nearly identical accuracy to full fine-tuning on the English XNLI test set. The LoRA model updates fewer than 1% of parameters yet matches the performance of the fully fine-tuned baseline from Task 2.

This result demonstrates that English NLI requires only modest adjustments to GPT-2’s pre-trained representations. LoRA’s low-rank adapters are sufficient to capture these adjustments without modifying the full parameter set.

Method	Params	Acc. (%)
Full FT (baseline)	124M	78.58
LoRA (ours)	1.18M (0.94%)	79.20
Full FT (best LR)	124M	81.56

Table 5: Comparison of fine-tuning strategies on XNLI (English test set).

4.5 Extension 2 — Unified Multilingual GPT-2 LoRA

Figure 2 compares four multilingual adaptation strategies across five target languages (English, German, Spanish, French, Swahili): (i) English-only full FT (Task 3), (ii) per-language full FT (Task 3), (iii) unified multilingual full FT (Task 3), and (iv) unified multilingual LoRA model.

LoRA achieves the highest accuracy for English among the multilingual models, avoiding the degradation seen in unified full FT. It also outperforms unified full FT on all non-English languages, showing that low-rank updates provide a more stable optimisation path than updating the full parameter set. Finally, LoRA comes close to matching the per-language full FT models, despite requiring only a single model rather than five separate ones.

These results indicate that LoRA is a more reliable and efficient approach to multilingual fine-tuning for GPT-2, although the overall performance ceiling remains limited by GPT-2’s monolingual pretraining.

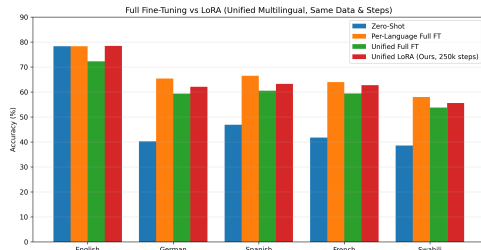


Figure 2: Comparison of English-only full FT, per-language full FT, unified full FT, and unified LoRA on GPT-2 across five languages.

4.6 Extension 3 — English Qwen2–1.5B LoRA and Zero-Shot Cross-Lingual Transfer

LoRA fine-tuning on Qwen2–1.5B produces strong zero-shot transfer despite training only on English XNLI. In Figure 3, the model reaches 70–90% accuracy on many XNLI languages and consistently outperforms GPT-2, whose zero-shot accuracy often remains below 40%.

This behaviour reflects the effect of multilingual pretraining: Qwen2 base already encodes cross-lingual semantics, allowing LoRA to adjust to the NLI task for other languages. In contrast, GPT-2’s monolingual training limits its ability to transfer the task. Notably, Qwen2–1.5B LoRA performs well on NLI in languages where it received no task-specific fine-tuning, demonstrating that parameter-efficient adaptation can exploit multilingual base to achieve broad cross-lingual generalisation.

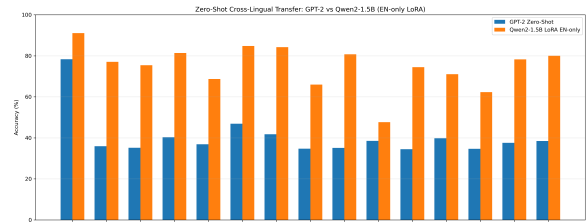


Figure 3: Zero-shot accuracy of Qwen2–1.5B English-only LoRA vs GPT-2 LoRA across 15 XNLI languages.

4.7 Extension 4: Unified Multilingual Qwen2–1.5B LoRA

Multilingual LoRA on Qwen2–1.5B improves performance across all non-English languages compared to the English-only LoRA model. As shown in Figure 4, German, Spanish, and French see small but consistent gains (+0.9 to +3.1 points), while Swahili improves substantially (+16.9 points). English accuracy drops slightly (−1.1 points) but remains high at 90%.

These results show that Qwen2’s multilingual pretraining enables LoRA to benefit from multilingual supervision without the interference effects seen in GPT-2. The strong improvement for Swahili also suggests that parameter-efficient tuning can meaningfully help lower-resource languages when built on top of a multilingual foundation.

5 Limitations

This work has several practical limitations that affect the comparability of results.

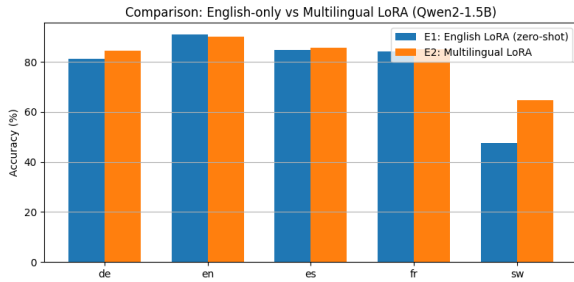


Figure 4: Qwen2–1.5B: English-only LoRA vs. multilingual LoRA across five languages.

First, the implementations of Task 2, Task 3, and the extension experiments were developed separately. Although the core training objectives and hyperparameters are aligned, the pipelines differ slightly in preprocessing and batching. As a result, comparisons across sections are indicative rather than fully controlled.

Second, unified multilingual training was constrained by compute cost. In Task 3 and Extension 2, only 50% of the dataset was used or training was capped at 250k steps. In Extension 4, we subsampled 50% of each language and trained for only one epoch. Ideally, all multilingual experiments would be rerun under a single consistent setup, but this was not feasible within time and resource limits.

Finally, we could not run full-parameter fine-tuning on Qwen2–1.5B. A direct comparison between full FT and LoRA at this scale would provide a more complete picture, but the required compute was prohibitive.

These constraints do not change the qualitative trends observed—LoRA’s efficiency, GPT-2’s multilingual ceiling, and Qwen2’s strong cross-lingual capacity—but they limit the precision of quantitative comparisons across experiments.

6 Conclusion

Across four extension experiments, we evaluated parameter-efficient fine-tuning and multilingual adaptation for GPT-2 and Qwen2–1.5B. LoRA consistently matched or exceeded full fine-tuning on English NLI while updating fewer than 1% of parameters, confirming that GPT-2 requires only small task-specific modifications. However, GPT-2’s monolingual pretraining imposes a fundamental ceiling: neither full FT nor LoRA substantially improves cross-lingual generalisation.

In contrast, Qwen2–1.5B exhibits strong zero-shot transfer after English-only LoRA training,

achieving competitive accuracy across many XNLI languages without target-language supervision. Multilingual LoRA further improves performance particularly for lower-resource languages such as Swahili, while retaining high English accuracy. These results show that the breadth and quality of the underlying multilingual pretraining, rather than the fine-tuning strategy itself, is the dominant factor governing cross-lingual NLI performance.

Overall, our findings demonstrate that LoRA is an efficient and robust alternative to full FT, but its effectiveness is ultimately shaped by the representational capacity of the base model. Future work should evaluate PEFT methods under fully controlled training pipelines, expand multilingual data coverage, and explore comparisons with full fine-tuning on large-scale models where compute constraints currently limit experimentation.

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